AMATH/PMATH 331: Applied Real Analysis Fall 2026

Assignment 3b Submission; due Tuesday, October 27, 2026

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[1] Let p be a monic polynomial of odd degree $2k+1$. This means that

$$p(x) = \sum_{j=0}^{2k+1} c_j x^j$$

with $c_j \in \mathbb{R}$ for $j \in \{0, 1, \ldots, 2k+1\}$ and $c_{2k+1} = 1$. Let $M = \max_{0 \le j \le 2k+1} |c_j| \ge 1$.

[a] Show that $p(2M) > 0$.

Hint: You will need to use the fact that $\sum_{j=0}^{2k} 2^j = \frac{1-2^{2k+1}}{1-2} = 2^{2k+1} - 1$.

Answer:

$$p(2M) = (2M)^{2k+1} + \sum_{j=0}^{2k} c_j (2M)^j$$

Now, let's bound the sum term:

$$|\sum_{j=0}^{2k} c_j (2M)^j| \le \sum_{j=0}^{2k} |c_j| (2M)^j$$

Since $(2M)^j > 0$ because $M \ge 1$.
But also we have that $|c_j| \le M$ for all j. So

$$|\sum_{j=0}^{2k} c_j (2M)^j| \le \sum_{j=0}^{2k} 2^j M^{j+1}$$

Now since $M \ge 1$ and $j \le 2k$, we have $M^{j+1} \le M^{2k+1}$. so

$$|\sum_{j=0}^{2k} c_j (2M)^j| \le \sum_{j=0}^{2k} 2^j M^{2k+1} = M^{2k+1}(\sum_{j=0}^{2k} 2^j)$$

$$|\sum_{j=0}^{2k} c_j (2M)^j| \le M^{2k+1}(2^{2k+1} - 1)$$

$$|\sum_{j=0}^{2k} c_j (2M)^j| \le (2M)^{2k+1} - M^{2k+1}$$

Since $M \ge 1$, $M^{2k+1} \ge 1$:

$$|\sum_{j=0}^{2k} c_j (2M)^j| < (2M)^{2k+1}$$

Therefore,

$$p(2M) = (2M)^{2k+1} + \sum_{j=0}^{2k} c_j (2M)^j > (2M)^{2k+1} - (2M)^{2k+1}$$

$$p(2M) > 0$$

[b] Show that $p(-2M) < 0$.

Answer:

$$\begin{aligned} p(-2M) &= (-2M)^{2k+1} + \sum_{j=0}^{2k} c_j(-2M)^j \\ &= \sum_{j=0}^{2k} c_j(-2M)^j - (2M)^{2k+1} \end{aligned}$$

Now, we can reuse the same bounds as above:

$$|\sum_{j=0}^{2k} c_j(-2M)^j| \leq \sum_{j=0}^{2k} |c_j|(2M)^j < (2M)^{2k+1}$$

so

$$\begin{aligned} p(-2M) &= \sum_{j=0}^{2k} c_j(-2M)^j - (2M)^{2k+1} \\ &< (2M)^{2k+1} - (2M)^{2k+1} \\ p(-2M) &< 0 \end{aligned}$$

[c] Prove that there exists at least one point $x_o \in \mathbb{R}$ such that $p(x_o) = 0$.

Answer:

Proof. p is a continuous map on $[-2M, 2M]$. Since $p(-2M) < 0 < p(2M)$, by the Intermediate Value Theorem, there exists $x_0 \in (-2M, 2M) \subset \mathbb{R}$ such that $p(x_0) = 0$. □

[2] Suppose that the surface of the Earth can be reasonably well-approximated to be a sphere $S = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = R\}$ for some $R > 0$. It is intuitively clear, and not hard to prove, that S is *path-connected*. You do not have to show this.

Let $T : S \to \mathbb{R}$ be the temperature function on the surface of the Earth. That is, $T(\mathbf{x})$ is the temperature at the point $\mathbf{x} \in S$. Assume that T is a *continuous* function. Show that there must exist at least one point $\mathbf{p}$ on the surface of the Earth which is at exactly the same temperature at the *antipodal* point $-\mathbf{p}$. That is, prove that there exists $\mathbf{p} \in S$ such that $T(\mathbf{p}) = T(-\mathbf{p})$.

Hint: Consider the function $f : S \to \mathbb{R}$ given by $f(\mathbf{x}) = T(\mathbf{x}) - T(-\mathbf{x})$.

Answer:

Proof. Consider the function $f : S \to \mathbb{R}$ given by $f(\mathbf{x}) = T(\mathbf{x}) - T(-\mathbf{x})$. Since T is continuous, f is also continuous on S.

We have to show that there exists $p \in S$ such that $f(p) = 0$.

Take arbitrary $x \in S$. Since S is path connected, there exists a path between x and $-x$. That is, there exists continuous map:

$$\gamma : [0, 1] \to S \quad \text{such that} \quad \gamma(0) = -x \quad \gamma(1) = x$$

Now consider $h = f \circ \gamma : [0, 1] \to \mathbb{R}$. Since f and γ are continuous, so is h.

$$h(0) = f(\gamma(0)) = f(-x) = T(-x) - T(x) = -f(x)$$

$$h(1) = f(\gamma(1)) = f(x)$$

If $f(x) = 0$, we are done. Otherwise, $-f(x)$ and $f(x)$ have opposite signs. Therefore 0 lies between $h(0)$ and $h(1)$. So by the Intermediate Value Theorem,

$$\exists \rho \in (0, 1) : h(\rho) = 0$$

Therefore, $f(\gamma(\rho)) = 0$. So let $p = \gamma(\rho) \in S$.

$$f(p) = 0$$

□

[3] Let $f : (0, \infty) \to \mathbb{R}$ be given by $f(x) = \frac{1}{x}$. We proved in class that f is continuous on $(0, \infty)$.

[a] Show that f is *not* uniformly continuous on $A = (0, 1)$.

Answer:
We have to show that:

$$\exists \varepsilon > 0 : \ \forall \delta > 0 : \exists x, y \in A : \ \|x - y\| < \delta \wedge \|f(x) - f(y)\| \geq \varepsilon$$

Proof. Let $\varepsilon \in (0, 1)$. Let arbitrary $\delta > 0$.
Let $x = \frac{1}{n}$ and $y = \frac{1}{n+1}$ for some $n \geq 2 \in \mathbb{N}$. (which are both in A)
Choose $n \geq 2$ suffieciently large such that $\frac{1}{n(n+1)} < \delta$.
Then,

$$|x - y| = \left| \frac{1}{n(n+1)} \right| = \frac{1}{n(n+1)} < \delta$$

But also

$$|f(x) - f(y)| = |n - (n+1)| = 1 > \varepsilon$$

So f is not uniformly continuous on A. □

[b] Show that f is uniformly continuous on $B = [1, \infty)$. [Note that B is *not compact*, so you cannot use the theorem from class. You need to do this directly.]

Answer:
We have to show that:

$$\forall \varepsilon > 0 : \ \exists \delta > 0 : \forall x, y \in B : \ \|x - y\| < \delta \to \|f(x) - f(y)\| < \varepsilon$$

Proof. Let arbitrary $\varepsilon > 0$. Let $\delta = \varepsilon$.
take arbitrary $x, y \in B$ such that $|x - y| < \delta$.

$$|f(x) - f(y)| = \left| \frac{y - x}{xy} \right| = \frac{|y - x|}{xy}$$

since $x, y \in B$, $xy \geq 1$ so $\frac{|y-x|}{xy} \leq |y - x|$. Thereofre:

$$|f(x) - f(y)| \leq |y - x| < \delta = \varepsilon$$

□

[4] Fix a subset $E \subseteq \mathbb{R}^n$. Let $f, g : E \to \mathbb{R}^m$.

[a] Suppose that f and g are both uniformly continuous on E. Let $\lambda, \mu \in \mathbb{R}$. Show that $\lambda f + \mu g$ is also uniformly continuous on E.
[This shows that the space of functions from E to $\mathbb{R}^m$ which are uniformly continuous on E is a linear subspace of the vector space of all continuous functions from E to $\mathbb{R}^m$.]

Answer:

Proof. Let aribtrary $\varepsilon > 0$. Let $\varepsilon' = \frac{\varepsilon}{|\lambda|+|\mu|}$.
Let δ_f and δ_g needed for the definition of uniform continuity of f and g
Let $\delta = \min \delta_f, \delta_g$
Take arbitrary $x, y \in E$ such that $\|x - y\| < \delta$. Since δ is the minimum of δ_f and δ_g:

$$\|x - y\| < \delta_f \qquad \|x - y\| < \delta_g$$

Therefore, by uniform continuity:

$$\|f(x) - f(y)\| < \varepsilon' \qquad \|g(x) - g(y)\| < \varepsilon'$$

Let $h = \lambda f + \mu g$ for brevity.

$$\begin{aligned}
\|h(x) - h(y)\| &= \|\lambda f(x) + \mu g(x) - \lambda f(y) - \mu g(y)\| \\
&= \|\lambda(f(x) - f(y)) + \mu(g(x) - g(y))\| \\
&\leq \|\lambda(f(x) - f(y)) + \|\mu(g(x) - g(y))\|\| && \text{By } \Delta \\
&\leq |\lambda|\|f(x) - f(y)\| + |\mu|\|g(x) - g(y)\| \\
&< |\lambda|\varepsilon' + |\mu|\varepsilon' = \varepsilon'(|\lambda| + |\mu|) = \varepsilon
\end{aligned}$$

So h is uniformly continuous on E. □

[b] Let $m = 1$, so $fg : E \to \mathbb{R}$ is defined. Suppose that f and g are both uniformly continuous on E, and that they are both *bounded*. Show fg is uniformly continuous on E.
[This shows that the space of *bounded* uniformly continuous functions from E to $\mathbb{R}$ is closed under multiplication. In the language of abstract algebra, it is a *subring* of the ring of continuous functions from E to $\mathbb{R}$.]

Answer:

Proof. Since f, g are bounded, there exists $M_f, M_g \in \mathbb{R}$ such That

$$|f(x)| \leq M_f, \quad |g(x)| \leq M_g \qquad \forall x \in E$$

Let arbitrary $\varepsilon > 0$ Let $\varepsilon_f = \frac{\varepsilon}{2M_f}$ and $\varepsilon_g = \frac{\varepsilon}{2M_g}$.
Let δ_f and δ_g needed for the definition of uniform continuity of f and g
Let $\delta = \min \delta_f, \delta_g$
Take arbitrary $x, y \in E$ such that $\|x - y\| < \delta$. Since δ is the minimum of δ_f and δ_g:

$$\|x - y\| < \delta_f \qquad \|x - y\| < \delta_g$$

Therefore, by uniform continuity:

$$\|f(x) - f(y)\| < \varepsilon_f \qquad \|g(x) - g(y)\| < \varepsilon_g$$

$$\begin{aligned}
|f(x)g(x) - f(y)g(y)| &= |f(x)g(x) - f(x)g(y) + f(x)g(y) - f(y)g(y)| \\
&= |f(x)(g(x) - g(y)) + g(y)(f(x) - f(y))| \\
&\leq |f(x)(g(x) - g(y))| + |g(y)(f(x) - f(y))| \qquad \text{By } \Delta \\
&\leq |f(x)||g(x) - g(y)| + |g(y)||f(x) - f(y)| \\
&\leq M_f|g(x) - g(y)| + M_g|f(x) - f(y)| \\
&< M_f\varepsilon_f + M_g\varepsilon_g \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon
\end{aligned}$$

□

[c] Show that the assumption that f, g are bounded is necessary for part [b]. That is, find examples of $f, g : E \subseteq \mathbb{R}^n \to \mathbb{R}$ which are uniformly continuous on E, and such that at least one of f, g is unbounded, and such that fg is not uniformly continuous on E. [Take $n = 1$ for simplicity.]

Answer:

Let $f, g : \mathbb{R} \to \mathbb{R}$ given by $f(x) = g(x) = x$.

Both are uniformly continuous on $\mathbb{R}$ and both are unbounded. However, $(fg)(x) = x^2$ is not uniformly continuous on $\mathbb{R}$ as we have shown in class.

[5] Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is continuous on $\mathbb{R}^n$ and that f *goes to zero at infinity* in the sense of problem [5] from Assignment #3a. Prove that f is uniformly continuous on $\mathbb{R}^n$.

Answer:

Proof. Since f goes to zero at infinity, we have that:

$$\forall \varepsilon > 0 : \ \exists M > 0 : \ \|x\| > M \to |f(x)| < \frac{\varepsilon}{2}$$

let arbitrary $\varepsilon > 0$.

Consider the closed ball $\overline{B_{M+1}(0)}$.

$$x \in \overline{B_{M+1}(0)} \iff \|x\| \leq M + 1$$

Since $\overline{B_{M+1}(0)}$ is compact and f is continuous on $\overline{B_{M+1}(0)} \subseteq \mathbb{R}^n$, we have that f is uniformly continuous on $\overline{B_{M+1}(0)}$. Therefore, there exists δ_0 such that

$$x, y \in \overline{B_{M+1}(0)}, \|x - y\| < \delta_0 \to |f(x) - f(y)| < \varepsilon$$

Now choose $\delta = \min(\delta_0, 1)$.

Take arbitrary $x, y \in \mathbb{R}^n$ such that $\|x - y\| < \delta$.

if both x, y are in $\overline{B_{M+1}(0)}$, we are done by compactness. Otherwise, at least one of them is outside the closed ball. So without loss of generality, assume:

$$\|x\| > M + 1$$

Now:

$$\begin{aligned} &\|x\| - \|y\| \leq \|x - y\| < \delta \leq 1 \\ &\|x\| - \|y\| < 1 \\ &\|y\| > \|x\| - 1 > M + 1 - 1 \\ &\|y\| > M \end{aligned}$$

So both points are outside radius M. Therefore,

$$|f(x)| < \frac{\varepsilon}{2} \qquad |f(y)| < \frac{\varepsilon}{2}$$

So

$$|f(x) - f(y)| \leq |f(x)| + |f(y)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

□